

Analysis Of Reddit Data using Large Language Models:

By: Kotagiri Sragvi

Introduction:

Reddit is a platform on the internet which allows users to post and comment on different subjects and fields related. This report explains the implementation of the problem statement, "Reddit data collection and LLM based text analysis" with the objective of this project being: "learn and use unfamiliar API, collect and structure real-world data and apply modern NLP techniques using LLM APIs".

The project goes according to the flow:

Data Collection -> Data Structuring -> Data classification -> Data Analysis -> Final Report

The sections below explain the implementation of this project.

Implementation:

For ease of understanding, the implementation process has been divided into:

- > Data Collection
- > Data Processing
- > Data Analysis
- > Analyzed Data Representation

Before executing or starting this project, make sure to have the libraries mentioned in "requirements.txt" installed and ready to be used

Data Collection:

For collecting data from reddit, reddit used to provide an official reddit API which was free to be created and accessed by developers world wide. Recently, in late 2026, reddit changed its API usage and enabled strict walls and permissions to create a reddit API. This makes it hard for new developers to create a reddit API and use it since the approval process takes weeks to be completed. Hence, PRAW (Python Reddit API Wrapper) library could not be used in this project. In this process of creating a reddit API, my initial account was banned by reddit.

Although reddit has forced these harsh conditions, it has also launched another website named as "Devvit" (developers.reddit.com) which allows users to launch apps which can only be interacted with in reddit's environment. This does not allow the app to be accessed and used for extracting data from reddit using external codes and programs. This causes us to use "Node.js" in our project for the data extraction. Following the below steps will allow you to collect data from reddit directly, printed in the console.

NOTE: If you have existing data, you can skip this section.

- > Go to developers.reddit.com and create a new app, the bare, web app.
- > Open CMD in your device and connect it with your devvit account (npm run login).
- > Execute the commands shown on the devvit website to finish creating your app.
- > Open the app folder locally and add reddit's permission as true in the file "devvit.json".
- > Next, open the file "index.ts" in the directory "/src/server" and paste the code written in the text file "index.ts.txt" in the project folder.
- > Now open CMD in the device and run the app (npm run devvit playtest <name_of_subreddit_created_by_default>)
- > Click on create a new post, wait for a few seconds and go back to the console (CMD), you shall find the data (about 100 posts and 500 comments) sitting in there.
- > Save the entire console's data as "<subreddit_name>-data-console.txt".

Following the above steps will help you in getting raw, real-world, trending data directly from reddit without any hindrance. I have selected the subreddits "Technology", "Smartphones" and "Gaming" as they all have a common theme and are related to Computer Science and Engineering in a way. In the initial extraction, 100 posts and 572 comments from technology, 100 posts and 272 comments from smartphones and 99 posts and 567 comments were taken. Using devvit's app doesn't require any authentication and security risks and has no API limits as it works only inside the reddit environment.

Now we need to process this data we have collected from reddit. In the project folder, you shall find a folder named "ConsoleData", which contains the raw data I extracted from reddit and a python program named "LogParser.py". This program converts the raw data's text file into a json file and removes all the excess console commands and lines making it ready to be processed/cleaned. The newly generated files from this program need to be moved into the main project directory or one can change the "DataCleanser.py" program to fetch the json files from another location in the device.

Data Cleaning / Processing:

After moving the files to the main directory, you can find a python program named "DataCleanser.py" in the main project directory which takes the raw data json files and not only converts them into csv files but also do:

- 1. Separates posts and comments of the different sub-reddits used.
- 2. Removes duplicate entries in both posts and comments.
- 3. Add missing values in required fields.
- 4. Delete "deleted" comments and posts.
- 5. Converts the timestamp extracted into a readable format.

This python program cleans the raw data which was extracted from the reddit website using a devvit app inside the console into csv files with the fields: "post ID", "subreddit", "title", "body/self-text", "author", "creation timestamp", "score/upvotes", "number of comments" and "URL/permalink" for posts and "comment ID", "parent post ID", "parent comment ID", "subreddit", "comment text", "author", "timestamp" and "score/upvotes" for comments. This helps in the LLM based analysis as csv files are easier to go through when compared to json files for LLMs.

POSTS FIELDS:	post ID	subreddit	title	body/self-text	author	creation timestamp	score/upvotes	number of comments	URL/permalink
About:	The integer value holding the post's ID given by reddit	The name of subreddit from which this post is taken	The title given to the post by the author		The creator of the post	The Epoch time when post was uploaded	Integer value holding the number of upvotes the post has received	Integer value of the number of comments	The link directed to the particular post
COMMENTS FIELDS:	comment ID	parent post ID	parent comment ID	subreddit	comment text	author	timestamp	score/upvotes	
About:	The integer value holding the comment's ID given by reddit	The integral ID value of the post in which the comment was made	The integral ID value of the comment to which this particular comment was made (if applicable)	The name of subreddit from which this comment was taken	The content that the comment states	The username of the creator of that particular comment	The Epoch time when the comment was made	Integer value holding the number of upvotes the comment has received	

Data Analysis:

The python code for analysing this collected data is stored in two files named "MasterAnalysis.py" and "TechnologyCommentsAnalysis.py" due to API rating limits and time taken to complete the analysis. First the execution of MasterAnalysis.py shall take place. After the complete analysis, if there is some hindrance when it comes to technology comments (which arises due to their large size and token usage), we can use the TechnologyCommentsAnalysis.py program separately.

For analysing the data, I have used groq for every single post and comment collected and gemini for all the posts and first 45 comments of every subreddit because of the lower API rating in gemini's API. The use of groq and gemini over hugging face is due to its popularity and also the precision of the two AI models. The higher API limits in groq allowed me to use it for analyzing all data collected. A very important feature to keep in mind is that gemini has a RPD (Request per Day) limit, which is set for only 20 requests.

The specific models used were “Gemini Flash 3.5” and “Grow: llama-3.1-8b-instant”. I used these models over the rest of them due to their API rating limits, accuracy and precision and popularity.

The prompts used for the two models are:

```
make_row_prompt(rows_data: (iter_)): 2+ usages
prompt = (
    "Perform a rigorous linguistic analysis on the following items. "
    "For each text row, accurately identify and return:\n"
    "1. sentiment: (Positive, Negative, Neutral)\n"
    "2. toxicity: (Low, Medium, High)\n"
    "3. topics: A short comma-separated list of primary core keywords.\n\n"
    "Return the response strictly inside a JSON object wrapper with a 'results' list array. "
    "Ensure every JSON object item preserves the mapping identifier string field exactly as provided.\n"
    "Example format:\n"
    "{\n"
    "  \"results\": [\n"
    "    {\n"
    "      \"id\": \"example_id\", \"sentiment\": \"Neutral\", \"toxicity\": \"Low\", \"topics\": \"tech, hardware\"\n"
    "    }\n"
    "  ]\n"
    "}\n\n"
    "Target payload inputs to scan:\n"
)
```

```
"Analyze the following social media text compilation layer corpus sample. "
"Provide a high-level metadata abstraction containing:\n"
"1. A structural 3-sentence summary highlighting core subject discussions, overall consensus tone, and user behavior.\n"
"2. Exactly 3 distinct prominent recurring linguistic conversation patterns observed in the content.\n\n"
"Return the response strictly as a structured JSON object containing keys 'summary' (string) "
"and 'patterns' (a list array of exactly 3 strings).\n"
"Example format:\n"
"{\n"
"  \"summary\": \"The text focuses primarily on... Users display a mixed response to... The primary consensus is...\",\n"
"  \"patterns\": [\n"
"    \"Frequent deployment of platform jargon.\",\n"
"    \"Sarcastic complaints regarding pricing strategies.\",\n"
"    \"Technical feature comparison matrices between generations.\"\n"
"  ]\n"
"}\n\n"
"Corpus Text Core Sample:\n{concatenated_text}"
```

This image on the left shows the prompt used for each individual comment, the same prompt is used for both groq and gemini and gives the sentiment analysis, topic and toxicity of the post or comment. The prompt on the right is used to get the summary and patterns in the respective subreddit's posts or comments, that is, every post and comment data of a specific subreddit (example: technology_posts.csv) has a separate summary and pattern recognised using the LLMs via this one prompt.

Every csv data file which was fetched and processed from reddit, example, Technology_posts.csv, Gaming_comments.csv, etc are analyzed in five different aspects and each data file input produces five csv formatted files which are generalized as:

1. {subreddit-name}_{posts/comments}_{AI-used}_Sentiment
2. {subreddit-name}_{posts/comments}_{AI-used}_Topics
3. {subreddit-name}_{posts/comments}_{AI-used}_Toxicity
4. {subreddit-name}_{posts/comments}_{AI-used}_Summarization
5. {subreddit-name}_{posts/comments}_{AI-used}_Patterns

Hence, we get an output of 60 csv files containing 5 different analysis methods using 2 different AI models from the 6 data files of posts and comments of 3 different subreddits around a common theme.

All the API calls and input and output from the AI models are documented in the file named “llm_execution_audit.csv” inside the folder “LLMAnalysis”. The 60 files (csv formatted) are also placed inside this folder to maintain low entropy in the main project folder/directory.

The python program used for this analysis also bears the rate limits and pauses the program when the limit is exceeded and we get a 403 http response from the API. The pause period is 30 seconds for both groq and gemini. The RPD established by gemini's models is the developer's responsibility (this means that I had to take care of RPD as we cannot pause the execution for 24 hours).

Results:

In this section, we shall discuss how precise the analysis was and also go through the visual representation of the LLM analysis. The graphs and plots we see in this section are a condensed form of the analyzed data, that is, the 60 csv files in the folder LLMAnalysis.

Below are a few examples of the comment text and their analysis according to groq and gemini.

1. This is a comment from the gaming subreddit and the analysis by the two AI models look reasonable, with gemini's sentiment looking more better than groq's.

```
GAMING COMMENT:
t1_or1e3yd,t3_1tz3yb5,t3_1tz3yb5,gaming,"I have 3 girlfriends (transbian polycule), and
since we're all long distance, I was thinking that we should try to find a f2p preferably
live service game to all play together, any recommendations?

2 of us already play warframe so I'm thinking about getting the other 2 into it.
",Unfair_Ad_598,2026-06-11 14:09:41,1

SENTIMENT ACCORDING TO GEMINI: positive
SENTIMENT ACCORDING TO GROQ: neutral

TOXICITY ACCORDING TO GEMINI: low
TOXICITY ACCORDING TO GROQ: low
```

2. This is another comment from the gaming subreddit with reasonable analysis from gemini but not from groq as the comment text is clearly positive.
3. This is a technology comment, in which we can see that the AI models disagree completely on the sentiment basis and toxicity basis. Gemini's response feels more reasonable given the context of this comment.

```
GAMING COMMENT:
t1_orxmwiw,t3_1tz3yb5,t3_1tz3yb5,gaming,"this is a great thread for
asking questions, thanks for posting!",CHjelmfort,2026-06-16
06:02:25,1

SENTIMENT ACCORDING TO GEMINI: positive
SENTIMENT ACCORDING TO GROQ: neutral

TOXICITY ACCORDING TO GEMINI: low
TOXICITY ACCORDING TO GROQ: low
```

```
TECHNOLOGY COMMENT:
t1_ovghs7s,t3_1umsz0n,t3_1umsz0n,technology,"I don't think people are
rejecting AI as much as they're rejecting the idea that they bear the
costs while someone else captures the value. If communities get more
noise, more water stress, and higher electricity demand, they naturally
ask: What's in it for us*?*",ExcellentBandicoot57,2026-07-04 05:50:14,2

SENTIMENT ACCORDING TO GEMINI: negative
SENTIMENT ACCORDING TO GROQ: positive

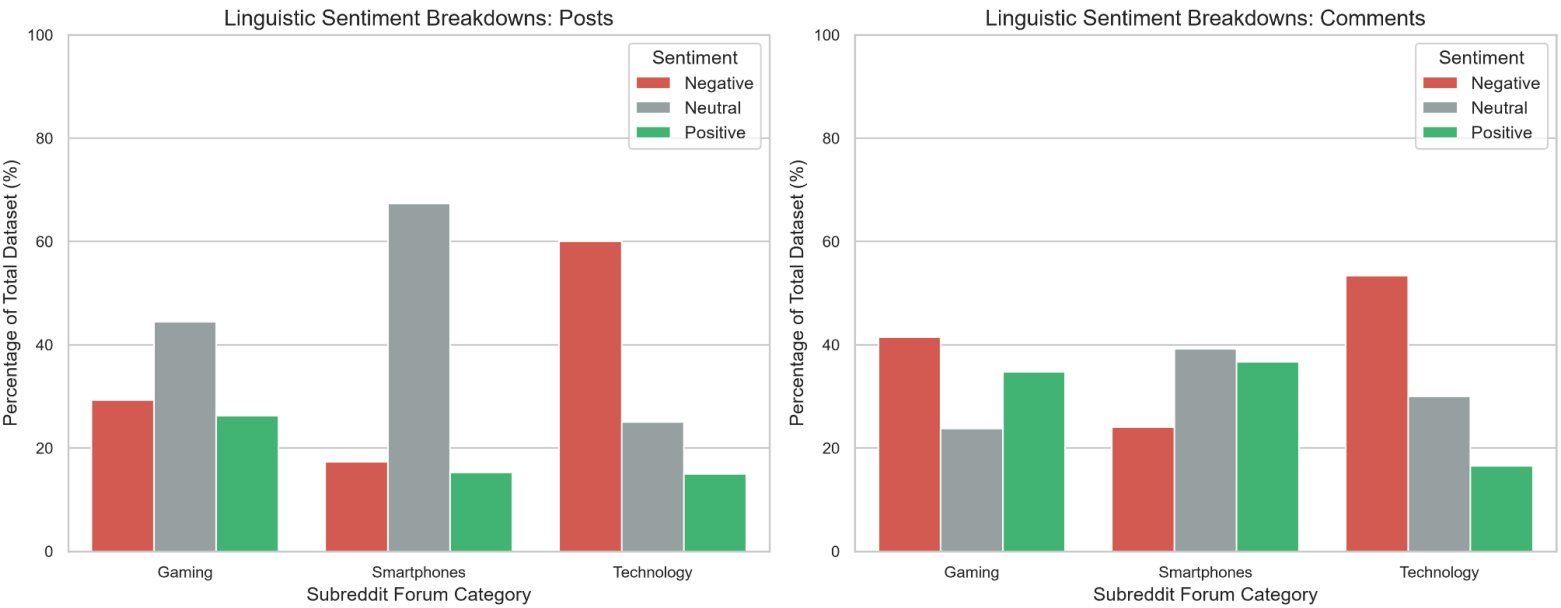
TOXICITY ACCORDING TO GEMINI: non-toxic
TOXICITY ACCORDING TO GROQ: toxic
```

We can conclude from the above results that gemini's analysis was more on point than groq's but they do not differ much when compared on a large scale. We used the gemini model for only the first 45 comments in any subreddit due to API restrictions and the comparison of the output by the two AI models for those 45 rows of data is given below:

LLM MODEL AGREEMENT AUDIT REPORT	
 Gaming Posts -> Sentiment Agreement Rate: 57.78% (Sample Window: 45 rows)	
 Gaming Posts -> Toxicity Agreement Rate: 77.78% (Sample Window: 45 rows)	
 Gaming Comments -> Sentiment Agreement Rate: 95.56% (Sample Window: 45 rows)	
 Gaming Comments -> Toxicity Agreement Rate: 100.00% (Sample Window: 45 rows)	
 Smartphones Posts -> Sentiment Agreement Rate: 68.89% (Sample Window: 45 rows)	
 Smartphones Posts -> Toxicity Agreement Rate: 93.33% (Sample Window: 45 rows)	
 Smartphones Comments -> Sentiment Agreement Rate: 60.00% (Sample Window: 45 rows)	
 Smartphones Comments -> Toxicity Agreement Rate: 86.67% (Sample Window: 45 rows)	
 Technology Posts -> Sentiment Agreement Rate: 73.33% (Sample Window: 45 rows)	
 Technology Posts -> Toxicity Agreement Rate: 66.67% (Sample Window: 45 rows)	

Most of the analyzed data is similar in both the models with the exception in the sentiment analysis of gaming posts, smartphones posts and smartphone comments and the toxicity of technology posts.

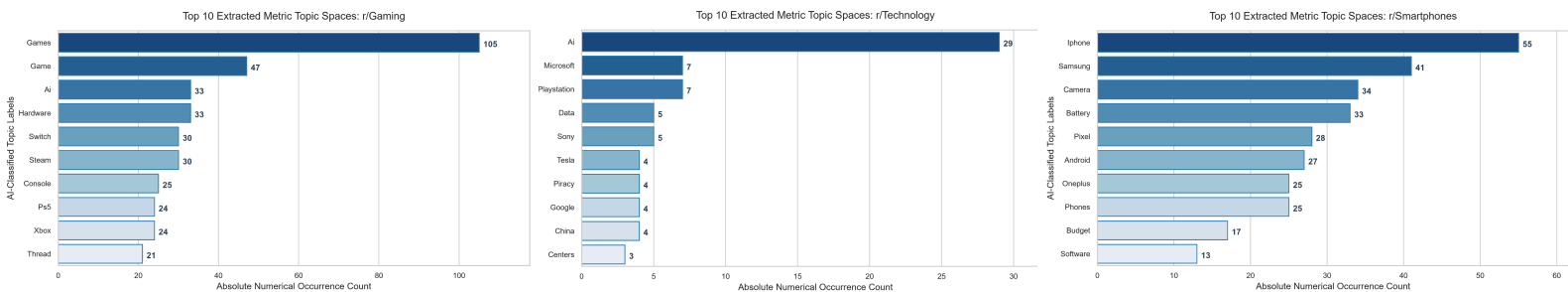
Linguistic Sentiment Profile Distributions Across Communities



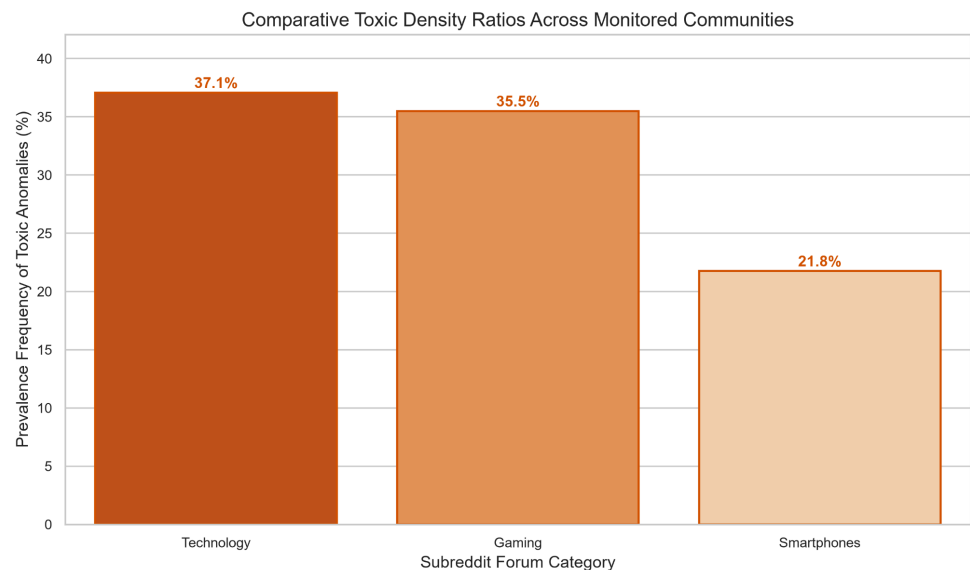
The above bar charts show us the sentiment distributions in the different subreddits and these bar charts are constructed on the basis of groq's output. With the name of the subreddit on the X-axis and percentage of sentiment on the Y-axis. This means that, in case of posts in gaming, the bars are divided according to the ratio in which they exist. Hence,

this graph shows and tells us about how diverse the subreddit communities can be, with smartphone posts being mostly neutral and technology posts being mostly negative. This graph also shows us that within gaming comments, there are more than 40% of those comments which are negative. This bar graph is also known as Sentiment Distribution.

The below graph shows us the topic frequency, this graph gives us information about which topic is trending in that particular subreddit and which topics are mostly frequently written by reddit users. The X-axis in all the three graphs is the absolute number of instances that topic has occurred and the Y-axis states the topic. The three graphs represent the three different subreddits we used for our project.



Another analysis we have done is for Toxicity, which is represented in the below bar graph. The graph shows us how toxic comments and posts are when compared among the three subreddits.



As we can see in the above graph, we can see that the toxicity of posts and comments of different subreddits are combined into one and the final result is shown in the form of a bar graph. The X-axis consists the names of the subreddit while the Y-axis shows the percentage of the number of times groq classified that particular row of data as toxic. From this graph we can say that the smartphones community is more friendly and non-toxic than the other two subreddits I selected. We see this difference in toxicity of the different subreddits because of the use of more slang terms and phrases by gamers and other tech enthusiasts.

A detailed report consisting the summarization and pattern recognition is present in the LLMvisuals folder, named as "subreddit_narrative_comparision.pdf".

Discussion:

Although this approach was good, it was not the best. The API rate limiting in AI models made me wait for more than 4 hours to get the outputs I wished. The use of devvit's app did not make me feel satisfied as I wanted to use reddit's official API but couldn't. The extra program files (like "TechnologyCommentsAnalysis.py" and "NarrativeReportGeneration.py") made this project a bit more complex to be explained.

The LLM outputs, according to me, are quite acceptable. We know that google's gemini flash-3.5 is the latest gemini model and is trained on millions of datasets and groq's llama-3.1-8b-instant model is also a fast developing AI model. The assurance is the compatibility between the two models, that is, the percentage showing how many outputs by the two models is more than 70% for most of the data (image in the results section). Hence, it is safe to conclude that these results are reliable and acceptable for the given data set.

The only ethical considerations I have is about the user anonymity and digital privacy as all that information has been given to the two API models without any anonymity. If I had more time, I would add this feature, anonymity, to the API calls so that the reddit users are protected. Also, with more time I might be able to gemini for all the data we have instead of analysing only the first 45 comments and skipping the rest in every subreddit. I would also improve the project directory, make it more clean and understandable.

Finally, I would like to state the use of LLM in this project. I used only one AI model, gemini's flash 3.5-extended for coding purposes and to help me understand the working of APIs. It was my idea and execution of devvit's app as an alternative for reddit's API with credentials. Gemini built the code in the specific way I asked it to, like pause the execution for at least 30 seconds when there is a 403 error, show the progress of the LLM analysis in the console so that I can know that the program is running successfully. These different prompts have given me a finalised "MasterAnalysis" python program consisting of 472 lines of code but still got stuck at technology comments. Hence, I created a separate program specifically for technology comments.

I have added pictures of the final project directory for ease of understanding below:

